

Statistical Modeling

DATA.STAT.740

Lecture 9

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Ordinal regression (generalized vector-GLM)

If the outcome variable is categorical, but **categories are ordered** we can use ordinal logistic regression

$$p_{i,k} = P(\mathbf{y}_i = k)$$

$$P(\mathbf{y}_i \leq k) = \sum_{j=1}^k p_{i,j} \quad (0 < P(\mathbf{y}_i \leq 1) < P(\mathbf{y}_i \leq 2) < \dots < P(\mathbf{y}_i \leq K) = 1)$$

Likelihood is the same for than for multinomial regression

$$l(\beta) = \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(p_{i,k})$$

Ordinal regression (generalized vector-GLM)

Systematic component - common linear predictor $\eta_i = x_i^T \beta$

Common slope assumption

No intercept term in x_i

Link function applies to cumulative probabilities

$$g(\text{P}(\mathbf{y}_i \leq k | x_i)) = \alpha_k - x_i^T \beta, \quad k = 1, \dots, K - 1$$

$$\alpha_1 < \alpha_2 < \dots < \alpha_{K-1}$$

Only thresholds change across categories - the effect of each predictor is the same regardless of which threshold you consider

Often we use proportional odds $g(\mu) = \log \left(\frac{\mu}{1 - \mu} \right)$

$$\log \left(\frac{\text{P}(\mathbf{y}_i \leq k | x_i)}{\text{P}(\mathbf{y}_i > k | x_i)} \right) = \alpha_k - x_i^T \beta$$

Ordinal regression (generalized vector-GLM)

Inverse link function

Cumulative probabilities

In this case this is the logistic CDF

$$P(\mathbf{y}_i \leq k | x_i) = \text{logit}^{-1}(\alpha_k - x_i^T \beta)$$

Category probability $p_{i,j} = P(\mathbf{y}_i = k) = \text{logit}^{-1}(\alpha_k - x_i^T \beta) - \text{logit}^{-1}(\alpha_{k-1} - x_i^T \beta)$

Likelihood (with thresholds α written explicitly as a parameter)

$$l(\beta, \alpha) = \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(p_{i,k})$$

$\log \left(\frac{P(\mathbf{y} \leq k | x)}{P(\mathbf{y} > k | x)} \right)$ decreases by β_j per unit increase in x_j
 $\beta_j > 0$: increasing x_j shifts probability mass toward lower categories

Proportional odds assumption

To use the presented formulation of the ordinal regression, you need to assume the proportional odds assumption (you should test the assumption)

For example: outline for the nominal-effects model test

1. proportional-odds model

$$\text{logit}(P(\mathbf{y} \leq k \mid x)) = \alpha_k - x^T \beta$$

2. nominal-effects model (slopes differ by category)

$$\text{logit}(P(\mathbf{y} \leq k \mid x)) = \alpha_k - x^T \beta_k, k = 1, \dots, K - 1$$

Likelihood-ratio test $H_0 : \beta_1 = \beta_2 = \dots = \beta_{K-1}$

H_A : At least one β_k differs

R-example!

Example: customer survey

Dataset: customersurvey.txt

A company has large stores in Helsinki and Tampere. The company conducts a customer satisfactory survey. At the store exits customers were able to express their satisfaction by the following type of survey:

Kuinka tyytyväinen olit asiakaspalveluumme?



Negative binomial regression

Poisson regression model is often used when observations are count data

Poisson model has a single parameter that defines the mean and the variance

$$\text{Var}(\mathbf{y}_i) = \text{E}(\mathbf{y}_i)$$

In real-world the count data often varies more than the Poisson model

$$\text{Var}(\mathbf{y}_i) > \text{E}(\mathbf{y}_i)$$

Negative binomial regression model handles the larger variance $\text{Var}(\mathbf{y}_i) = \mu_i + \frac{\mu_i^2}{\theta}$ with a **precision** parameter θ

Negative binomial model

Negative binomial model is

$$\mathbf{y}_i \sim \text{NegBinom}(\mu_i, \theta) = f(y_i \mid \mu_i, \theta) = \frac{\Gamma(y_i + \theta)}{\Gamma(\theta) y_i!} \left(\frac{\theta}{\theta + \mu_i} \right)^\theta \left(\frac{\mu_i}{\theta + \mu_i} \right)^{y_i}, \quad y_i = 0, 1, 2, \dots$$

$$\mu_i = \mathbb{E}(\mathbf{y}_i)$$

$$\text{Var}(\mathbf{y}_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

$$\eta_i = x_i^T \beta$$

$\eta_i = \log(\mu_i)$ NB has a canonical link $\log \left(\frac{\mu_i}{\mu_i + \theta} \right)$ but it's not used in practice

R-example!

Zero-inflated Poisson model

In count data models, there may be situations where the number of zero occurs significantly more frequently than it should occur under Poisson or negative binomial distribution.

Zero-inflated models are one way to model data are to have more frequent zeros

The zero-inflated Poisson model is a mixture model with two zero generating processes:

- The first process generates zeros

- The second process is a Poisson distribution

Zero-inflated Poisson model

The zero-inflated Poisson model has the probability structure

$$P(\mathbf{y}_i = 0) = \theta_i + (1 - \theta_i) \exp(-\mu_i)$$

$$P(\mathbf{y}_i = y_i) = (1 - \theta_i) \frac{\mu_i^{y_i} \exp(-\mu_i)}{y_i!}, \quad y_i = 1, 2, \dots,$$

consisting of degenerate and poisson components

most often the log-link is used for the observation mean

$$\log(\mu_i) = x_i^T \beta$$

in addition, the mixture parameter θ_i can be modeled to be dependent on the predictors

$$\text{logit}(\theta_i) = x_i^T \alpha$$

R-example!

Quasi-Models

Quasi-likelihood models extend GLM framework to situations where

you have a good idea of the the mean-variance structure

you're not sure about the exact outcome distribution

not supported by any of the previously discussed models

Quasi-likelihood Q is defined by

$$\frac{\partial Q}{\partial \mu_i} = \frac{y_i - \mu_i}{\phi V(\mu_i)}$$

Defined by the mean and the variance

Solution to the associated score equations give $\hat{\beta}$ but Q is not a real likelihood

Quasi-Models

Quasi-GLMs and associated score equations are compatible with IRLS algorithm (available in R function glm)

Quasi-GLMs require less assumptions about the outcome distribution

We get

- consistent estimates of regression coefficients,
- robust standard errors, even if the distribution is misspecified.

Drawbacks (due to lack of the likelihood function)

AIC/BIC are not available

Likelihood ratio tests are invalid.

Quasi-Models

In practice, we use quasi-models when there is **overdispersion**

observed variance > model variance

You can use visual diagnostics (Patterns or excessive scatter → overdispersion)

Pearson residuals vs fitted values

Deviance residuals vs fitted values

Plot squared Pearson residuals

You can also diagnose for overdispersion

$$\hat{\phi} = \frac{\text{Residual Deviance}}{N - p}$$

$\hat{\phi} \approx 1.0$ no overdispersion

$\hat{\phi} > 1.5$ Likely overdispersion

$\hat{\phi} > 2.0$ Serious overdispersion

Quasi-Binomial regression

Quasi-binomial regression is a **quasi-likelihood GLM for modeling binary outcomes** when data exhibit overdispersion or underdispersion (relative to the standard binomial model)

In Binomial regression, the observations are have a Binomial distribution

$$\mathbf{y}_i \sim \text{Binom}(m_i, p_i) = \binom{m_i}{y_i} p_i^{y_i} (1 - p_i)^{m_i - y_i}$$

$$E(\mathbf{y}_i) = m_i p_i$$

$$\text{Var}(\mathbf{y}_i) = m_i p_i (1 - p_i)$$

In quasi-Binomial regression, we model

$$\mu_i = m_i p_i$$

$$\text{logit}(p_i) = x_i^T \beta$$

$$\text{Var}(\mathbf{y}_i) = \phi m_i p_i (1 - p_i) \quad \text{R-example!}$$

Quasi-Poisson regression

Quasi-Poisson regression is a generalized linear model for count data that:

- Uses the same mean structure as a Poisson regression

- Uses the same log link

- BUT** allows the variance to be larger than the mean (**overdispersion**)

- Similar to negative-binomial regression

Poisson: $E(\mathbf{y}_i) = \text{Var}(\mathbf{y}_i) = \mu_i, \quad \mathbf{y}_i \sim \text{Poisson}(\mu_i)$

Quasi-Poisson: $E(\mathbf{y}_i) = \mu_i, \quad \text{Var}(\mathbf{y}_i) = \phi\mu_i$

Likely overdispersed : Use quasi-Poisson model

Serious overdispersion : Use negative-Binomial model

R-example!

End of Lecture 9